

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 2_Q8

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

A bank generates secure codes using 3-digit numbers where each digit is unique, and the code must be divisible by 3. You are tasked with generating the first N such codes based on user input, ensuring the digits are unique and the number is divisible by 3.

Note: Use nested for loops to solve.

Input Format

The first line contains an integer N representing the number of valid codes to generate.

Output Format

The output prints N lines, each line contains a valid 3-digit code.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

Output: 102

105

108

120

123

Answer

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();

        int count = 0;

        // Loop for hundreds digit (1-9) to ensure 3-digit number
        for (int i = 1; i <= 9 && count < N; i++) {
            // Loop for tens digit (0-9)
            for (int j = 0; j <= 9 && count < N; j++) {
                // Loop for units digit (0-9)
                for (int k = 0; k <= 9 && count < N; k++) {
                    // Check unique digits
                    if (i != j && j != k && i != k) {
                        int num = i * 100 + j * 10 + k;
                        // Check divisibility by 3
                        if (num % 3 == 0) {
                            System.out.print(num + " ");
                            count++;
                        }
                    }
                }
            }
        }
    }
}
```

}
}

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Status : Correct

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Marks : 10/10

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